TT-F-322D
October 24, 1973
SUPERSEDING
Int. Fed. Spec. TT-F-00322C (GSA-FSS)
December 9, 1971 and
Fed. Spec. TT-F-322B
April 12, 1970

FEDERAL SPECIFICATION

FILLER TWO-COMPONENT TYPE: FOR DENTS, CRACKS, SMALL-HOLES, AND BLOW HOLES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two-component type filler, resin and hardener or curing agent, mixed in proportion recommended by manufacturer for use in filling dents, cracks, small holes, etc. The filler may be applied by trowel or spatula.

1.2 Classification.

1.2.1 Type. The filler shall be of one type, a blend resin and a curing agent or hardener.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-E-489 - Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces).
TT-S-735 - Standard Test Fluids; Hydrocarbon.
PPP-B-636- Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gauge and Lighter.
PPP-F-320- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 141/GEN. - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specification, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 2240 - Hardness, Indentation.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

National Motor Freight Classification.

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P. Street, N. W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Composition. The filler is a blend of components A and B, to be furnished in two separate containers.

3.2.1 Component A. In paste form, consisting of commercially available synthetic resin with suitable extender or pigment.

3.1.2 Component B. A curing agent consisting of benzoyl peroxide with suitable extender and necessary amount of plasticizers shall be furnished in cream or paste form.

3.1.2 Ingredients. The ingredients of the filler, components A and B blended together in accordance with industry's commercial practice shall be as specified in table I.

TABLE I. Components A and B

Ingredients	Requirements	
	Minimum	Maximum
Component A:		
Pigment, percent by weight of component A	60	66
Resin content, percent by weight of component A	34	40
Component B:		
Paste or cream		
Benzoyl peroxide, percent by weight of component B	48	--
Extender, percent by weight of component B	--	52

3.1.4 Color. If color of component B is specified in the contract or purchase order (see 6.2) it shall be an approximate match of a color chip in Fed. Std. 595.

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3.2 Quantitative requirements.

3.2.1 When components A and B are mixed in accordance with manufacturer's direction the product shall comply to all requirements specified in Table II.

TABLE II. Quantitative requirements of the mixture

Characteristics	Minimum	Maximum
Gel time, minutes @ 77 F.	4.5	6.5
Time to attain shore D-20, minutes	7.0	11.0
Time to attain shore D-40, minutes	13.0	20.0
Time to attain shore D-60, minutes	18.0	25.0
Hardness	55	75
Adhesion psi	1000	----
Viscosity, cps RVT Brookfield-TD Spindle, 2.5 RPM	400,000	600,000

NOTE: Run the viscosity within 4 minutes after mixing.

3.3 Qualitative requirements.

3.3.1 Condition in container. A freshly opened full container of components A and B, shall be free from lumps, skins, thickening, and shall show no more settling or caking than can be readily reincorporated to a smooth, homogeneous state when tested as in 4.3.3.

3.3.2 Storage stability. Component A when tested as in 4.3.4 shall show no skinning, lumps or hard settling or caking.

3.3.3 Working properties. The filler, when tested as in 4.3.5 shall spread smoothly and shall show no tendency to gouge.

3.3.4 Hydrocarbon resistance. When tested as in 4.3.6 the filler shall show no lifting, blistering, wrinkling, or softening immediately upon removal from the test. After 24 hours air-drying at room temperature, the portion that was immersed shall be almost indistinguishable from the portion that was not immersed.

3.3.5 Water resistance. The filler, when tested as in 4.3.7 shall show no lifting, blistering, wrinkling, or softening immediately upon removal from the test. After 24 hours air-drying at room temperature, the portion that was immersed shall be almost indistinguishable from the portion that was not immersed.

3.3.6 Resistance to lifting with top coat. When tested as in 4.3.8 the filler shall resist lifting during application of a top coat. After 24 hours air-drying of the top coat the dried film shall show no irregularities such as lifting, mottling, peeling, or orange peel, etc.

3.3.7 Shrinkage. When tested as in 4.3.16, the filler shall show no cracking and no more than 1.0 mm shrinkage away from the rim.

3.3.8 Impact test. A film of the filler shall show no cracking, flaking, or loss of adhesion to the test panel when tested as in 4.3.14.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Test Methods.

4.3.1 The test shall be conducted as indicated in table II.

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TABLE III. Index

	Requirement Reference	Applicable Tests	
		Fed. Test Method Std. No. 141	Paragraph Reference
Color	3.1.4	----	4.3.2
Condition	3.3.1	3011	4.3.3
Storage stability	3.3.2	----	4.3.4
Working properties	3.3.3	----	4.3.5
Hydrocarbon resistance	3.3.4	----	4.3.6
Water resistance	3.3.5	----	4.3.7
Resistance to lifting	3.3.6	----	4.3.8
Component A:			
Pigment content	Table I	4021	-----
Resin content	Table I	----	4.3.9
Component B:			
Benzoyl peroxide content	Table I	----	4.3.10
Extender content	Table I	----	4.3.11
Curing time	Table II	----	4.3.12
Adhesion	Table II	----	4.3.13
Impact	3.3.8	----	4.3.14
Hardness	Table II	----	4.3.15
Shrinkage	3.3.7	----	4.3.16

4.3.2 Color. Spread about 20 grams of sample (component B, paste) on a clear glass panel (3- by 5-inch) to about half the area. Use spatula to smooth and even the surface and allow to dry for 24 hours at room temperature. Compare the color visually against the standard color chip for compliance with 3.1.4.

4.3.3 Condition in container. Determine the condition in the container in accordance with method 3011, paragraph 4.1.3 of Fed. Test Method Std. No. 141 except delete the last part of the sentence "of satisfactory brushing consistency."

4.3.4 Storage stability. Place three ounces of component A in a four-ounce glass jar, close tightly, place in an oven at 132°F. for 24 hours. At the end of this period allow the materials to condition for 2 hours at room temperature. Then observe for compliance with 3.3.2.

4.3.5 Working properties. Prepare 3 steel panels (3- by 6-inch) by sanding the surfaces lightly with No. 80 grit paper, wiping off any loose particles then cleaning with solvent using petroleum naptha-ethylene glycol monoethyl either mixture. On each of the three cleaned panels apply approximately 0.030 inch film thick of the mixed filler (see 1.1) and allow to dry at room temperature for 24 hours.

4.3.6 Hydrocarbon resistance. Immerse one of the test panels prepared in 4.3.5 in a test fluid conforming to TT-S-735, type III for 4 hours in accordance with method 6011 of Fed. Test Method No. 141. At the end of the test period remove and observe for compliance with 3.3.4 Air dry the panel at room temperature for another 24 hours and again examine for compliance with 3.3.4.

4.3.7 Water resistance. Immerse another of the test panel prepared in 4.3.5 in distilled water maintaining the temperature at 23° ± 1°C. for 18 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. At the end of the test period remove and examine the film for compliance with 3.3.5. Air dry the panel for 24 hours and again examine for compliance with 3.3.5.

4.3.8 Resistance to lifting with top coat. Brush a coat (approximately 1 mil dry film) enamel conforming to TT-E-489 on the other test panel prepared in 4.3.5. Allow the top coat to dry at room temperature for 24 hours then observe for compliance with 3.3.6.

4.3.9 Resin content. Determine the resin content of component A by calculation using the percent pigment obtained by method 4021, paragraph 3.7 of Fed. Test Method Std. No. 141. Based on 100 percent total solids, substract percent pigment from 100 equals percent of resin.

4.3.10 Peroxide content.

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Reagents

1. Acetone
2. 40 percent NaI in water solution
3. Glacial acetic acid
4. 0. 1 N Sodium thiosulfate solution

Procedure

Into a 500 ml. glass-stoppered Erlenmeyer flask add 100 ml. acetone. Vent the flask with a glass thread which has been sterilized in a Bensen burner flame. Weigh accurately on analytical balance the proper amount of peroxide to be analyzed.

Suggested weights:

Cream or Paste Hardener 0.7 to 0.8 g

Saturate the acetone with small pieces of dry ice. Add the weighed sample. Swirl to dissolve the sample. Then add more dry ice to maintain Co_2 atmosphere. Add 9 ml. of 40 percent sodium iodide solution, followed by 1 ml. of glacial acetic acid. Stir for 3 minutes. Add 50 ml. of distilled water and allow mixture to stand 3 more minutes. Titrate with 0.1 sodium thiosulfate solution to a pale yellow, add starch indicator and continue titration to a colorless end point.

Calculations

$$\text{Active Oxygen} = \frac{(\text{ml thiosulfate}) \times (\text{N. thiosulfate}) \times (800)}{\text{Weight sample (mg)}}$$

$$\text{Percent Benzoyl Peroxide} = \frac{(\text{Active Oxygen}) \times (100)}{6.6}$$

4.3.11 Extender content. Determine extender content of the curing agent or hardener by difference as follows: 100 percent minus benzyol peroxide (obtained from paragraph 4.3.10) equals percent extender. Observe for compliance with table I.

NOTE: All components of the curing agent or hardener are soluble with same solvent.

4.3.12 Curing (gel) time. When components A and B are mixed in accordance with manufacturer's direction it shall cure as specified in table II.

4.3.13 Adhesion. This test is made by bonding together two overlapping steel panels (1- by 3-inch) prepared as in 4.3.5. *Apply the mixed filler as in 4.3.5 one square inch on both strips,* then press together, squeeze and allow to cure for 24 hours at room temperature. After 24 hours the strips are pulled apart by means of tensile testing machine, the reading taken at failure and converted to psi shall comply with the requirements of table II.

4.3.14 Impact test. Apply approximately a 0.050 inch film thick of filler on a 4- by 12-inch No. 20 gage steel panel solvent cleaned using petroleum naphtha-ethylene glycol monoethyl ether mixture. Allow to dry at room temperature for 24 hours and place on a block of wood with the coated side up. Drop a 5 pound steel ball from a height of 18 inches on the center of the film. Evaluate for compliance with 3.3.8.

4.3.15 Hardness (Identation). Determine the hardness of the cured (hard) filler in accordance with ASTM D 2240 (by means of a Durometer D scale) for compliance with table II.

4.3.16 Shrinkage. Solvent clean two small shallow tin or terneplate lids approximately two inches in diameter and three-eights inch high. Fill with the filler and level with top of the lids. *Care should be taken to work out all bubbles. Allow to dry for 7 days at room temperature* then subject the specimens to the following cycles:

16 hours at minus 40°F
8 hours at plus 150°F

Repeat this cycle for four times and determine for compliance with 3.3.7.

4.3.17 Inspection of preparation for delivery. The filler components in separate containers shall be examined for compliance with packaging, packing, and marking requirements of section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The component A material shall be packaged in 1-quart or 1-gallon cans conforming to PPP-C-96, type V, class A. The component B material shall be packaged in containers of the type normally used for the product.

5.1.2 Level B. The filler shall be packaged in accordance with paragraph 5.1.1 except that the component A cans shall not require the side seam coating.

5.1.3 Level C. Unless otherwise specified, the filler shall be packaged in accordance with the manufacturers standard practice providing that it insures protection for the product during shipment and safe delivery to its destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Four gallons or 12 quart cans of the component A material shall be placed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be provided with egg-erate type partitions constructed from material conforming to PPP-F-320, class weather-resistant, to separate the cans. One container of the compound B material shall be placed within each cell of the box. Closure and sealing of the boxes shall be in accordance with the appendix to PPP-B-636.

5.2.2 Level B. The filler shall be packed in accordance with paragraph 5.2.1 except that the boxes shall comply with PPP-B-636, class domestic and the partitions shall conform to PPP-F-320, class - domestic.

5.2.3 Level C. The filler packaged as specified in 5.1.3 shall be packed in containers that will insure carrier acceptance and at destination in compliance with the Uniform Freight Classification or the National Motor Freight Classification.

5.3 Marking. Marking shall be in accordance with 5.3.1 and 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The metal filler is intended primarily for filling dents and small holes in truck body and automotive repair work. It is also satisfactory for repairing flow-holes, sand holes and surface blemishes in metal castings. The filler is not recommended for use on aluminum or magnesium surfaces.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Size of container (see 5.1).
(c) Level of packaging, level of packing, and marking (see 5.1 and 5.2).
(d) Color, if required (If using agency desire specific color) see 3.1.4.

MILITARY CUSTODIANS:

Army -MR DOT, Navy
Air Force - 85

User Activities:

Air Force - 84
Navy -SH

CIVIL AGENCIES COORDINATING ACTIVITIES:

DOT, FAA
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POS

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6279

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